

SOFTWARE ENGINEERING

Software Requirements Specifications (SRS)

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Problem Statement: Chat Application (Socket Programming)

Team Number: 5

Revision History

Version	Date	Author	Change Summary	Approval
1.0	05/09/2026	Sakshi Srinivas Ghodke Yashita Anand Ujwal K Varun Sahu	SRS with diagrams embedded	

Approvals

Role	Name	Signature/Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for a Real-Time Chat Application built using socket programming. It defines the client-server communication model, message exchange protocols, and interfaces, intended as a reference for developers and evaluators.

1.2 Scope

Covers a multi-client chat system where users connect to a central server over TCP sockets to exchange one-to-one and group messages in real time, including presence tracking, file sharing, and basic moderation. Excludes third-party messaging protocol integration (e.g., XMPP, SIP) and mobile push-notification infrastructure.

1.3 Audience

Developers, QA Engineers, System Integrators, and Assessment Evaluators.

1.4 Definitions

TCP, IP, Socket, Port, Thread, Client, Server, ACK (Acknowledgement), UI, JSON, SSL/TLS, GUI.

2. Overall description

2.1 Product perspective

The Chat Application is a client-server system. A central multi-threaded server listens on a fixed port and accepts multiple concurrent client socket connections. Each connected client runs a lightweight desktop/console/GUI application that sends and receives messages through a persistent TCP socket connection to the server, which routes messages to intended recipients or rooms.

2.2 Major product functions (detailed)

- User registration and login (username/password authentication)
- One-to-one (private) messaging
- Group chat / chat room creation and management
- Broadcast messaging (server-wide announcements)
- File/attachment transfer between users
- Online/offline presence and user list display
- Typing indicators
- Message delivery and read acknowledgements
- Chat history storage and retrieval
- Administrative moderation (kick/ban/mute users)
- Graceful reconnection handling on network drop

2.3 User roles and characteristics (expanded)

- **User (Client):** general chat participant; expects low-latency delivery and a simple interface.
- **Administrator/Moderator:** manages rooms, can mute/kick/ban users, broadcasts announcements.
- **Server Operator:** monitors server health, connection counts, and logs.

2.4 Operating environment

Server runs on a host machine (Linux/Windows) with a fixed listening port over TCP/IP; clients run on desktop OS with network access to the server (LAN or WAN). Requires a persistent socket connection maintained via keep-alive/heartbeat.

2.5 Constraints

Must use TCP sockets (not HTTP polling) for real-time delivery; single-threaded-per-client or thread-pool model on the server; message size limits for buffer handling; optional TLS wrapping for encrypted transport.

3. External interface requirements

3.1 User interfaces

Primary UI: a chat window with contact/room list, message pane, text input box, file-attach button, and online-status indicators. Accessible via keyboard navigation.

3.2 Hardware interfaces

No dedicated hardware; standard network interface card (NIC) for TCP/IP communication.

3.3 Software interfaces

- Server Socket API (ServerSocket/Socket, TCP) for client-server communication
- Local file system interface for chat log persistence and file transfer storage
- Optional database interface (e.g., SQLite/MySQL) for user credentials and message history

3.4 Communications

- TCP sockets over a configurable port (default 5000)
- Custom application-layer message protocol (JSON or delimited text) framed over the socket stream
- Optional TLS/SSL socket wrapper for encrypted communication
- Heartbeat/ping-pong messages every N seconds to detect dropped connections; automatic client reconnection with exponential backoff

4. System features (detailed)

IDs follow CHAT-F-####. Each requirement includes acceptance criteria and a test case reference.

4.1 Authentication & Session Management

Req ID	Requirement	Type	Priority	Source/ Stakeholder	Acceptance criteria/ Test	Comments
CHAT-F-001	The system shall allow a user to register a unique username and password before connecting to chat.	Functional	High	End User	AC: Duplicate usernames rejected with error. Test: TC-Auth-01	Requires user store
CHAT-F-002	The system shall authenticate a client's credentials over the socket connection before granting chat access.	Functional	High	Security	AC: Invalid credentials close the socket with an error message. Test: TC-Auth-02	
CHAT-F-003	The system shall terminate a session and close the socket cleanly on user logout.	Functional	High	End User	AC: Server removes client from active list within 1s of logout. Test: TC-Auth-03	
CHAT-F-004	The system shall detect abrupt client disconnection (socket exception/tim eout) and update presence accordingly.	Functional	High	Operations	AC: User marked offline within heartbeat interval after disconnect. Test: TC-Auth-04	Depends on heartbeat

4.2 Messaging

Req ID	Requirement	Type	Priority	Source/ Stakeholder	Acceptance criteria/ Test	Comments
CHAT-F-010	The system shall allow a client to send a private text message to another connected user by username/ID.	Functional	High	End User	AC: Message delivered to correct recipient socket only. Test: TC-Msg-01	
CHAT-F-011	The system shall deliver messages to online recipients in real time ($\leq$ NFR latency bound).	Functional	High	End User	AC: Message appears at recipient client without polling/refresh . Test: TC-Msg-02	
CHAT-F-012	The system shall queue messages for offline users and deliver them upon next login.	Functional	Medium	End User	AC: Queued messages delivered in original order after reconnect. Test: TC-Msg-03	Requires persistence
CHAT-F-013	The system shall send a delivery acknowledgement back to the sender once a message reaches the recipient.	Functional	Medium	End User	AC: Sender UI shows "delivered" tick. Test: TC-Msg-04	
CHAT-F-014	The system shall broadcast a typing indicator event to the relevant recipient/roo	Functional	Low	End User	AC: Typing indicator appears/disappears within 2s of state change. Test: TC-Msg-05	

	m when a user is composing a message.					
CHAT-F-015	The system shall persist all sent messages with sender, receiver, timestamp, and content to chat history storage.	Functional	High	Compliance	AC: History retrievable in correct order after restart. Test: TC-Msg-06	

4.3 Group Chat/Rooms

Req ID	Requirement	Type	Priority	Source/ Stakeholder	Acceptance criteria/ Test	Comments
CHAT-F-020	The system shall allow a user to create a named chat room/group.	Functional	High	End User	AC: Room appears in directory immediately. Test: TC-Grp-01	
CHAT-F-021	The system shall allow users to join or leave an existing chat room.	Functional	High	End User	AC: Membership list updates for all room participants. Test: TC-Grp-02	
CHAT-F-022	The system shall broadcast a message sent to a room to all currently connected members of that room.	Functional	High	End User	AC: All online members receive the message within latency bound. Test: TC-Grp-03	
CHAT-F-023	The system shall display the list of currently online users, globally and per room.	Functional	Medium	End User	AC: List refreshes on connect/disconnect events. Test: TC-Grp-04	

4.4 File Transfer

Req ID	Requirement	Type	Priority	Source/ Stakeholder	Acceptance criteria/ Test	Comments
CHAT-F-030	The system shall allow a user to send a file attachment (up to a configured size limit) to another user or room over the socket connection.	Functional	Medium	End User	AC: File received intact (checksum match) at recipient. Test: TC-File-01	Buffer chunking
CHAT-F-031	The system shall reject file transfers exceeding the configured maximum size with an error notice to the sender.	Functional	Medium	Operations	AC: Oversized file rejected before transfer starts. Test: TC-File-02	

4.5 Administration & Moderation

Req ID	Requirement	Type	Priority	Source/ Stakeholder	Acceptance criteria/ Test	Comments
CHAT-F-040	The system shall allow an administrator to broadcast a system-wide announcement to all connected clients.	Functional	Medium	Admin	AC: All active sockets receive the announcement within latency bound. Test: TC-Admin-01	
CHAT-F-041	The system shall allow an administrator to kick or ban a user from a room or the server, closing	Functional	High	Admin	AC: Banned user's subsequent connection attempts are refused. Test: TC-Admin-02	

	their socket connection.					
CHAT-F-042	The system shall log all administrative actions (kick/ban/broadcast) with timestamp and admin ID for audit.	Functional	Medium	Compliance	AC: Log entries created for every admin action. Test: TC-Admin-03	

5. Non-Functional requirements (detailed)

IDs follow CHAT-NF-####.

Req ID	Requirement	Category	Priority	Acceptance criteria/ Measurement
CHAT-NF-001	The system shall deliver chat messages with an end-to-end latency of ≤ 500 ms for at least 95% of messages under normal operating conditions (≤ 100 concurrent clients).	Performance	High	Load test shows 95th percentile message latency ≤ 500 ms. Test: TC-Perf-01
CHAT-NF-002	The server shall support at least 200 concurrent client socket connections without dropping active sessions.	Scalability	High	Load test with 200 simulated clients maintains stable connections. Test: TC-Scale-01
CHAT-NF-003	The server shall automatically detect and clean up dead/orphaned socket connections within 30 seconds via heartbeat timeout.	Reliability	High	Simulated abrupt disconnect is detected and cleaned within 30s. Test: TC-Rel-01
CHAT-NF-004	Chat message history shall be retained and queryable for at least 90 days.	Data Retention	Medium	Historical messages older than today but within 90 days are retrievable. Test: TC-Ops-01
CHAT-NF-005	The client UI shall remain responsive (input not blocked) while socket I/O operations are in progress, using asynchronous/non-	Usability	Medium	UI responsiveness verified during simulated slow network. Test: TC-UX-01

	blocking I/O or a separate listener thread.			
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5.1 Security

5.1.1 Security Objectives

- Protect user credentials and message content from interception during transit.
- Prevent unauthorized users from joining rooms or impersonating other users.
- Ensure only authenticated, authorized administrators can perform moderation actions.

5.1.2 Security Requirements

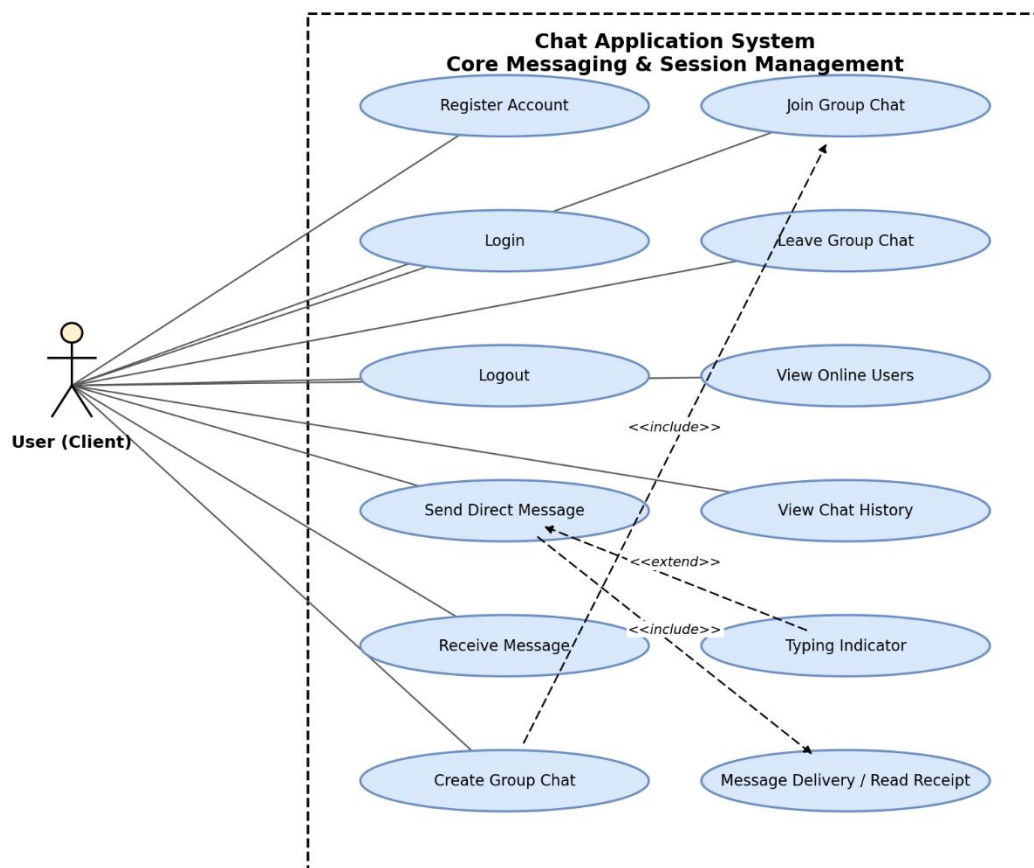
Req ID	Requirement	Type	Priority	Acceptance criteria/ Measurement
CHAT-SR-001	All socket communication shall be encrypted using TLS/SSL where deployed over untrusted networks.	Security	High	TLS handshake verified via packet capture. Test: TC-Sec-01
CHAT-SR-002	Passwords shall never be transmitted or stored in plaintext; passwords shall be hashed (e.g., bcrypt/SHA-256 with salt).	Security	High	DB inspection confirms hashed storage. Test: TC-Sec-02
CHAT-SR-003	The server shall validate and sanitize all incoming message payloads to prevent injection or buffer-overflow attacks.	Security	High	Fuzz test with malformed payloads does not crash server. Test: TC-Sec-03
CHAT-SR-004	The server shall enforce a session token per authenticated connection; messages without a valid token shall be rejected.	Security	High	Requests with missing/invalid token are refused. Test: TC-Sec-04
CHAT-SR-005	The server shall rate-limit connection attempts per IP to mitigate brute-force login and denial-of-service attempts.	Security	Medium	Repeated rapid login attempts trigger throttling. Test: TC-Sec-05

6. Quality attributes & Acceptance tests

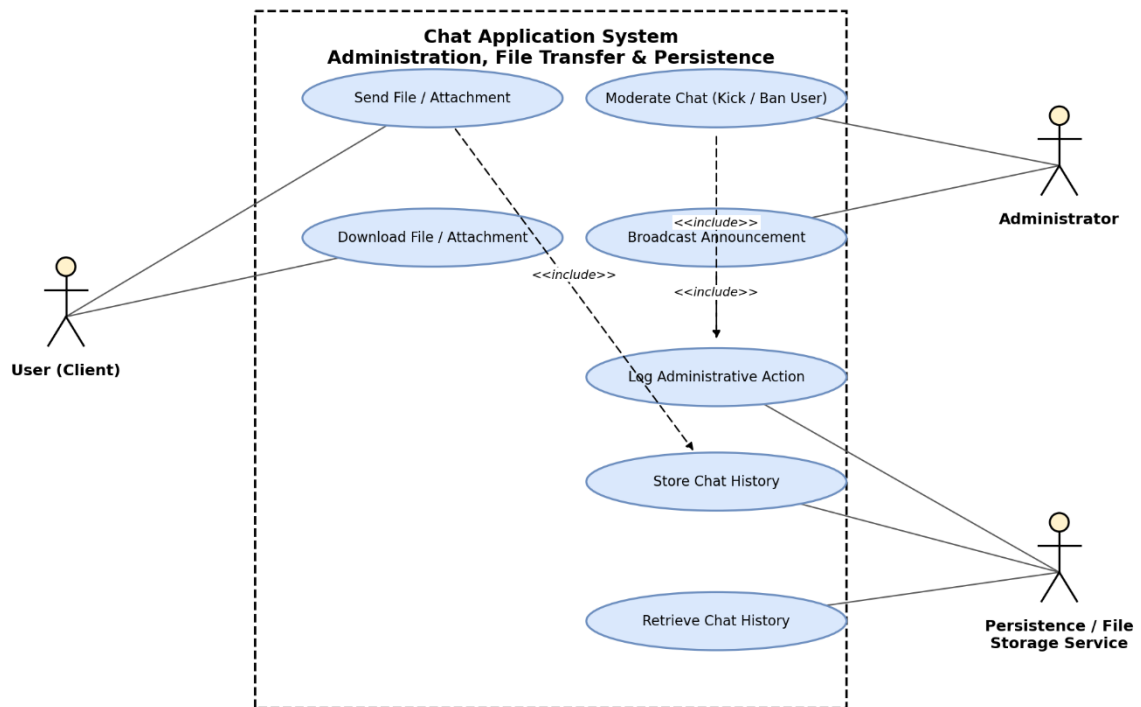
- **Exit criteria for acceptance:** All high-priority functional requirements implemented and verified; no critical NFR failures; RTM shows all test cases passed.
- **Acceptance test suites:** Authentication, Private Messaging, Group Chat, File Transfer, Administration/Moderation, Performance & Load, Security, Reliability/Reconnection.

7. UML Use-Case Diagram

Use Case Diagram 1 — Core Messaging & Session Management



Use Case Diagram 2 — Administration, File Transfer & Persistence



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement (short)	Section ref	Module	Test case(s)	Status
CHAT-F-001	Register account	4.1	AuthModule	TC-Auth-01	N
CHAT-F-010	Send private message	4.2	MessageRouter	TC-Msg-01	N
CHAT-F-020	Create chat room	4.3	RoomManager	TC-Grp-01	N
CHAT-F-030	Send file attachment	4.4	FileTransferModule	TC-File-01	N
CHAT-F-041	Kick/ban user	4.5	AdminModule	TC-Admin-02	N
CHAT-NF-001	Message latency target	5	MessageRouter	TC-Perf-01	N
CHAT-SR-001	TLS encryption	5.1.2	ConnectionManager	TC-Sec-01	N